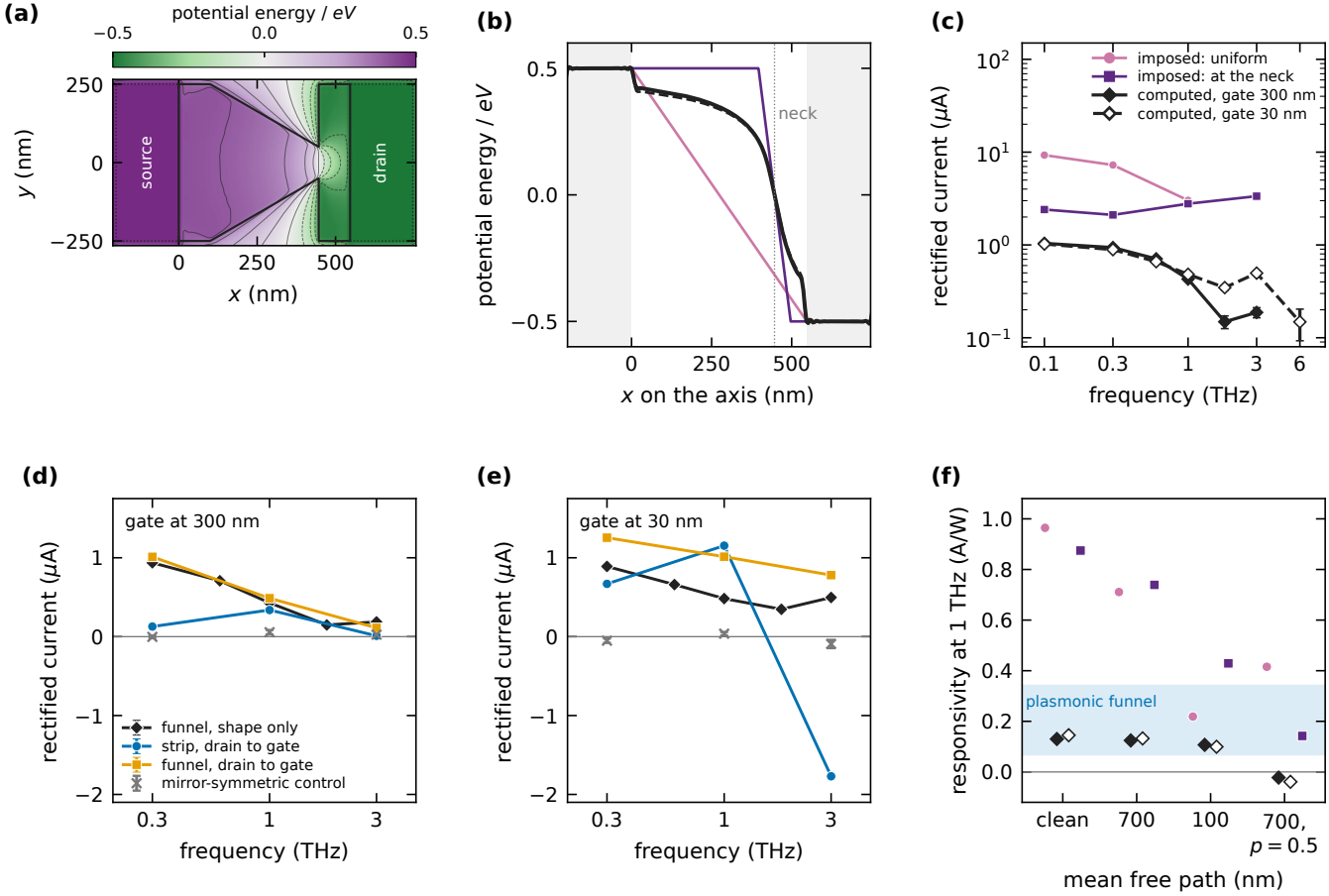

(a)
potential energy / eV
−0.5
0.0
0.5
250
0
−250
y (nm)
source
drain
0
250
500
x (nm)
(b)
potential energy / eV
0.5
0.0
−0.5
neck
0
250
500
x on the axis (nm)
(c)
imposed: uniform
imposed: at the neck
computed, gate 300 nm
computed, gate 30 nm
rectified current (μA)
10^2
10^1
10^0
10^−1
0.1
0.3
1
3
6
frequency (THz)
(d)
gate at 300 nm
rectified current (μA)
1
0
−1
−2
funnel, shape only
strip, drain to gate
funnel, drain to gate
mirror-symmetric control
0.3
1
3
frequency (THz)
(e)
gate at 30 nm
rectified current (μA)
1
0
−1
−2
0.3
1
3
frequency (THz)
(f)
responsivity at 1 THz (A/W)
1.0
0.8
0.6
0.4
0.2
0.0
plasmonic funnel
clean
700
100
700, p = 0.5
mean free path (nm)